16/7/26

EXP 1: BASIC NETWORKING COMMANDS IN LINUX AND WINDOWS

AIM:-

To study and execute basic networking commands available in linux and window operating system for diagnosing and trouble shooting network issue.

REQUIREMENTS:

Pc with windows/Linux OS, Terminal or command prompt, Active network connection.

THEORY / BACKGROUND:

A computer network is a collection of interconnected devices that share resources and information Networking commands are built-in utilities that allow administrators and user to configure, diagnose, and troubleshoot network connections.

Windows provides command such as ipconfig, ping tracert, netstat, nslookup, arp and route. Linux provides equivalents ipconfig / ip, ping, traceroute, netstat/ss, nslookup / dig, arp and route / ip route.

PROCEDURE:

1. Open command prompt (Windows) or Terminal (Linux)
2. Execute 'ipconfig' (windows) or 'ipconfig /ipa' (Linux) to view IP configuration
3. Use 'ping 8.8.8.8' to test connectivity to Google's DNS Server
4. Execute tracert google.com (windows)

```
[s]@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:12:f1:cc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86069sec preferred_lft 86069sec
    inet6 fe80::a00:27ff:fe12:f1cc/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[s]@localhost ~]$
```

```
[s]@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[s]@localhost ~]$
```

```
[s]@localhost ~]$ hostname
localhost.localdomain
[s]@localhost ~]$
```

```
[sj@localhost ~]$ ping google.com
PING google.com (142.251.43.142) 56(84) bytes of data.
64 bytes from bkk03s01-in-f14.1e100.net (142.251.43.142): icmp_seq=1 ttl=118 time=6.27 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=2 ttl=118 time=7.93 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=3 ttl=118 time=11.1 ms
64 bytes from bkk03s01-in-f14.1e100.net (142.251.43.142): icmp_seq=4 ttl=118 time=5.11 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=5 ttl=118 time=12.3 ms
```

```
[sj@localhost ~]$ nslookup google.com
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.205.110
Name:   google.com
Address: 2404:6800:4807:815::200e

[sj@localhost ~]$
```

```
[sj@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp6       0      0 localhost:ipp           [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
udp        0      0 localhost.locald:bootpc _gateway:bootps         ESTABLISHED
udp        0      0 0.0.0.0:mdns            0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp6       0      0 [::]:mdns               [::]:*
udp6       0      0 localhost:323           [::]:*
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7
Active UNIX domain sockets (servers and established)
```

```
[sj@localhost ~]$ arp -a
_gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3
[sj@localhost ~]$
```

```
[sj@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
qlen 1000
    link/ether 08:00:27:12:f1:cc brd ff:ff:ff:ff:ff:ff
[s]@localhost ~]$
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
qlen 1000
    link/ether 08:00:27:1f:d8:cc brd ff:ff:ff:ff:ff:ff
[sj@localhost ~]$
[sj@localhost ~]$ ss -tuln
Netid       State       Recv-Q       Send-Q                Local Address:Port                Peer Address:Port
udp         ESTAB       0            0                     10.0.2.15%enp0s3:68                     10.0.2.2:67
[sj@localhost ~]$
```

```
[sj@localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
```

or traceroute google.com' to view packet path.

5. Run 'netstat -an' to list active network connection and listening ports.

6. Use 'nslookup google.com' to perform DNS lookup.

7. Execute 'arp -a' to view the ARP cache.

8. Use 'route print' (Windows) or 'route -n' (Linux) to view the routing table

9. Record all outputs and analyze the results.

COMMANDS / CODE

```
ipconfig/all        #Windows - Full ip configuration
ip a                #Linux - IP address info
ping -c4 8.8.8.8    #Linux - ping 4 packets
ping 8.8.8.8
tracert 8.8.8.8     #Linux - Trace route
netstat -tuln       #Linux - Listening ports
nslookup google.com #DNS resolution
arp -a              #ARP table
route -n            #Routing table (Linux)
```

RESULT:

Basic networking commands were successfully executed on both Linux and Windows. IP addresses, routing table, ARP caches, and DNS resolution were observed and recorded.